

acm forum

[For policy on letters, see October 1970 issue]

Can ACM Serve All?

In his ACM President's Letter in the February 1972 issue of *Communications*,¹ Walter Carlson has by ellipsis misquoted the U.S. Department of Labor, and so caused Computer Science academicians much unnecessary grief.

The Department of Labor said²

"At the present time the computer sciences are not generally recognized by colleges and universities as a bona fide academic discipline with *standardized licensing, certification, or registration procedures.*" (Italics supplied.)

By omitting the words italicized above, President Carlson changed the meaning quite significantly. With those words in place, the statement is far more appropriate to the purposes of the Department of Labor, and seems quite true, if a bit ungrammatical.

Computer Science is not recognized as an academic *profession*, for there is no special disciplinary accreditation of Computer Science curricula, as there is for engineering, pharmacy, law, and other professional curricula. In most colleges and universities, Computer Science is a nonprofessional discipline in the School of Arts and Sciences, or in the Graduate School. Left to its own devices, academic Computer Science will undoubtedly opt to remain in the relative freedom of the Arts and Sciences, rather than be constrained to a professional mold, but that is another subject.

The spokesman for the Department of Labor was slightly muddled, in any case, to confuse academic Computer Science with preparation specifically for jobs in programming or systems analysis, but that confusion is widely held. A Computer Science major today, whether at bachelor or master level, is being prepared

not just to program, or to analyze real world problems in preparation for programming, but rather:

- to improve the tools that programmers and systems analysts use,
- to develop better machine systems, better software systems, better languages for communicating with the machines, and better methods for solving intractable problems,
- to manage activities that are strongly computer dependent,
- to teach about computers at college level, or
- to advance the fundamental theory of digital information processors.

Some graduates of Computer Science programs may be programmers or systems analysts, but they will have changed direction, or missed their aim, or simply still be in their learning phase.

How then does this affect ACM? The essential question is whether ACM is the Society of the computer scientists, or of the programmers and systems analysts, or of both. To be both is a greater challenge, and has significant financial and operational advantages, but requires a continuing recognition on the part of ACM that there are marked differences between the two camps, and that each must be served well if the Association is to prosper in the future as it has in the past.

Howard E. Tompkins
Computer Science Department
Indiana University of
Pennsylvania
Indiana, PA 15701

¹ Carlson, Walter M., "The Answer Is . . . Competence, Integrity, and Productivity . . . What is the Question?", *Comm. ACM*, 15 (Feb. 1972), 69.

² "Relative to Status of Data Processing Employees," *Federal Register*, Dec. 2, 1971; reprinted in relevant parts in *Comm. ACM* 15 (Feb. 1972), 129.

Discuss Impact of Computer on Society at Annual Meetings?

On Kent Curtis' point,¹ which seems to be that discussion of the implications and consequences of computing at ACM annual meetings is not appropriate to a professional society, I would say that this view is very widely, and very unfortunately, held throughout the computing community and within the ACM itself. It seems to me that if computing is a profession, and if we are professionals, then we must come to grips, through discussions at annual meetings and in official publications and elsewhere, with problems of the impact of the computer on society. It is just this concern which distinguishes the professional from the technician, and the profession from the trade. In the words of Anthony Wedgwood Benn, sometime [British] Minister of Technology,

It is high time cyberneticists and Ph.Ds. in math, and programmers and systems analysts, and hardware and software experts, come out of their secret world. They must . . . begin assuming citizenship responsibilities for the consequences of what they are doing. They must also understand the implications of all they are planning for us. For it could be the most revolutionary onslaught ever made by any group of men on any established society.²

If we should have these concerns, it seems to me that our professional society should be a focal point for their expression, just as it is in the area of technical concerns.

T. D. C. Kuch
7554 Spring Lake Drive
Bethesda, MD 20034

¹ Kent K. Curtis, Criticisms of ACM Annual Meetings, *Comm. ACM* 15 (Jan. 1972), 54.

² Anthony Wedgwood Benn, Speech to the British Computer Society, Southampton, Sept. 25, 1967, quoted in Nigel Calder, *Technopolis*, p. 313 f. (Simon and Schuster, 1970).